

Supplementary Materials for CS4013/5013: Artificial Intelligence

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Topic: Idealized Search

An Example of Search Problem Formulation

Figure 1 show a partial map of OU campus. Suppose we want to find a path to from the Sarkeys Energy Center (A) to the OU Rugby Clubhouse (F) with the smallest mileage.

This can be viewed as a search problem i.e., from a set of candidate paths, we are searching for one with the smallest mileage. In theory, there are infinitely many candidate paths, which makes search challenging. In practice, we can simplify the problem by only searching from a finite set of candidate paths. These candidates can be generated as follows:

Step 1. Pick a finite number of locations, including the initial one A and target one F. In this example, let's pick A, B, C, D, E, F, G. (Note: how to pick locations often depend on the problem setting; a good heuristic is of course to pick locations that we believe are likely to be crossed by the target path.)

Step 2. Measure the mileage to *directly* travel from one picked location to another, if the travel is allowed. (Note: if we do not care about mileage but only care about the number of stops, simply set all mileages to 1.)

Step 3. Generate a graph, whose nodes are the picked locations and weights are the mileages between them. In this example, we will obtain the graph in Figure 1 (right).

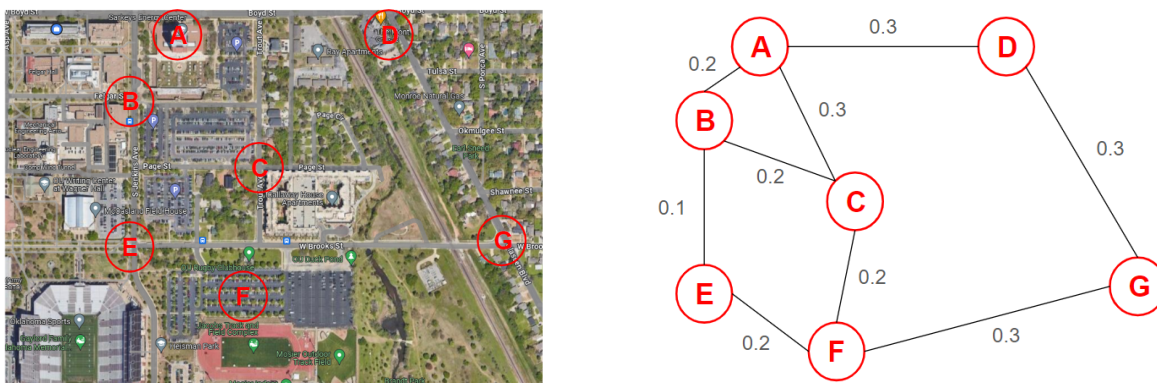


Figure 1: From Map Search (left) to Graph Search (right)

Our search problem is now simplified to: in the graph, find a path from node A to node F with the minimum total weight. We can then go ahead to apply different search